

What is a legacy system?

Ideal Answer

A legacy system is an old system that is valuable for the company which developed and owns it.

Q2

Why are legacy systems difficult?

Ideal Answer

Because they resist modification and evolution to meet changing business requirements.

Q3

What are the options available to manage legacy systems?

Ideal Answer

- Freeze
 - Outsource
 - Carry on maintenance
 - Discard and redevelop
 - Wrap
 - Migrate
-

Q4

What is wrapping?

Ideal Answer

Wrapping is a black-box modernization technique that surrounds the legacy system with a software layer hiding unwanted complexity.

Follow-up

Does wrapping change the internal system?

Expected Answer

No, it hides complexity using a software layer.

Q5

What is migration in legacy systems?

Ideal Answer

Migration moves an operational system to a new platform while retaining its functionality with minimal disruption.

Section 2 — Impact Analysis

دي متوقع جدًا يطلع منها شفوي

Q6

What is impact analysis?

Ideal Answer

Impact analysis estimates the parts of software affected by a proposed change request.

Q7

What are the classes of impact analysis techniques?

Ideal Answer

- Traceability analysis
 - Dependency analysis
-

Q8

What is traceability analysis?

Ideal Answer

It identifies related requirements, design, code, and test cases connected to the feature being changed.

Q9

What is dependency analysis?

Ideal Answer

It assesses the effects of change on dependencies between program entities.

Q10

What is ripple effect?

Ideal Answer

Ripple effect measures how likely a change in one module causes problems in the rest of the system.

Q11

What is change propagation?

Ideal Answer

It ensures that a change in one component is propagated properly throughout the system.

Section 3 — Refactoring

دي أكيدة شفوي

Q12

What is refactoring?

Ideal Answer

Refactoring changes the internal structure of software to make it easier to understand and cheaper to modify without changing observable behavior.

Q13

What are the goals of refactoring?

Ideal Answer

- Improve maintainability

- Remove duplicate code
 - Simplify code
 - Make code easier to understand
 - Add flexibility
-

Q14

Does refactoring change observable behavior?

Ideal Answer

No.

Section 4 — Program Comprehension

Q15

What is program comprehension?

Ideal Answer

It is the task of building mental models of a software system at different abstraction levels.

Q16

What is a mental model?

Ideal Answer

A mental model is the programmer's mental representation of the program.

Q17

What strategies are used in program comprehension?

Ideal Answer

- Bottom-up
- Top-down

- Opportunistic combination
-

Q18

What is chunking?

Ideal Answer

Chunking creates higher-level abstraction structures from lower-level structures.

Q19

What is cross-referencing?

Ideal Answer

Cross-referencing links elements from different abstraction levels.

Section 5 — Software Reuse

دي مهمة جداً.

Q20

What is software reuse?

Ideal Answer

Software reuse is using existing artifacts or software knowledge to build a new software system.

Q21

Who introduced software reuse?

Ideal Answer

Doug McIlroy.

Q22

What are the types of reusable artifacts?

Ideal Answer

- Data reuse
 - Architectural reuse
 - Design reuse
 - Program reuse
-

Q23

What are the characteristics of reusable components?

Ideal Answer

- Environmental independence
 - High cohesion
 - Loose coupling
 - Adaptability
 - Understandability
 - Reliability
 - Portability
-

Q24

What are the benefits of software reuse?

Ideal Answer

- Increased reliability
 - Reduced process risk
 - Increased productivity
 - Accelerated development
 - Improved maintainability
-

Section 6 — Maintenance Classification

دي أخطر جزء في المحاضرة ومتوقع جدًا.

سؤال متكرر جدًا

Q25

What are the intention-based types of maintenance?

Ideal Answer

- Corrective
- Adaptive
- Perfective
- Preventive

Q26

What is corrective maintenance?

Ideal Answer

Corrective maintenance fixes failures and defects in the software.

Q27

Why is corrective maintenance reactive?

Ideal Answer

Because it is performed after detecting defects.

Q28

What is adaptive maintenance?

Ideal Answer

Adaptive maintenance modifies software to adapt to environmental changes.

Q29

Give examples of adaptive maintenance.

Ideal Answer

- Supporting new hardware
 - Converting batch to online operation
 - Compatibility with other applications
-

Q30

What is perfective maintenance?

Ideal Answer

Perfective maintenance improves user experience, efficiency, and maintainability.

Q31

What is preventive maintenance?

Ideal Answer

Preventive maintenance prevents future problems by modifying software products.

Q32

What is software rejuvenation?

Ideal Answer

It is terminating the system, cleaning its internal state, and restarting it to prevent failures.

Section 7 — Evidence-based Classification

أو شفوي تصنيف MCQ الدكتوراة غالبًا هتجيب منه

Q33

Into how many clusters are maintenance activities grouped?

Ideal Answer

Four clusters.

Q34

What are the four clustering aspects?

Ideal Answer

- Whole software
 - External documentation
 - Program code properties
 - Customer functionality
-

Q35

What is the default maintenance type in ambiguous cases?

Ideal Answer

Evaluative.

Section 8 — Maintenance Organization

Q36

What are the maintenance activity types?

Ideal Answer

- Investigation
 - Modification
 - Management
 - Quality assurance
-

Q37

What are the organization-level maintenance elements?

Ideal Answer

- Event management
- Configuration management

- Change control system
-

Q38

Who approves change requests?

Ideal Answer

The change control board.

Q39

What is SLA?

Ideal Answer

A contract between customers and maintenance service providers specifying performance targets.

Q40

What are the three support levels?

Ideal Answer

- Level 1: Files reports
 - Level 2: Analyzes problems
 - Level 3: Performs actual software changes
-

الـ Trick Questions المتوقعة

Trick 1

Does refactoring change functionality?

Answer

No, only internal structure.

Trick 2

Which maintenance type is reactive?

Answer

Corrective maintenance.

Trick 3

Which maintenance type is called reengineering?

Answer

Perfective maintenance.

Trick 4

What is the default evidence-based type?

Answer

Evaluative.